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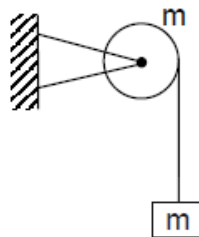
JEE Mains 2020 Jan Chapter wise Question Bank

Rotational Motion

Q1

A block of mass m is suspended from a pulley in form of a circular disc of mass m & radius R . The system is released from rest, find the angular velocity of disc when block has dropped by height h . (there is no slipping between string & pulley)

एक m द्रव्यमान का ब्लॉक धिरनी से रस्सी के द्वारा जुड़ा हुआ है। धिरनी चकती के रूप में है। जिसका द्रव्यमान m तथा त्रिज्या R है। यदि निकाय को विराम से छोड़ा जाता है। तब ब्लॉक द्वारा h ऊँचाई नीचे आने पर चकती का कोणीय वेग होगा (धिरनी तथा चकती में कोई फिसलन नहीं है)



- (1) $\frac{1}{R} \sqrt{\frac{4gh}{3}}$ (2) $\frac{1}{R} \sqrt{\frac{2gh}{3}}$ (3) $R \sqrt{\frac{2gh}{3}}$ (4) $R \sqrt{\frac{4gh}{3}}$

7th Jan Morning

Sol

(1)

$$mgh = \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2$$

$$v = \omega R \text{ (no slipping कोई फिसलन नहीं)}$$

$$mgh = \frac{1}{2}m\omega^2 R^2 + \frac{1}{2} \frac{mR^2}{2} \omega^2$$

$$mgh = \frac{3}{4}m\omega^2 R^2$$

$$\omega = \sqrt{\frac{4gh}{3R^2}} = \frac{1}{R} \sqrt{\frac{4gh}{3}}$$

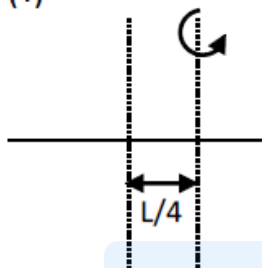
Q2

Rotational Motion**JEE Mains 2020 Jan Chapter wise Question Bank**

A thin uniform rod is of mass M and length L . Find the radius of gyration for rotation about an axis passing through a point at a distance of $\frac{L}{4}$ from centre and perpendicular to rod.

एक पतली एकसमान छड़ जिसका द्रव्यमान M और लम्बाई L है। तो छड़ के केन्द्र से $\frac{L}{4}$ दूरी और छड़ के लम्बवत् अक्ष के सापेक्ष घूर्णन त्रिज्या ज्ञात करें।

- (1) $\sqrt{\frac{7}{48}} L$ (2) $\sqrt{\frac{5}{48}} L$ (3) $\sqrt{\frac{7}{24}} L$ (4) $\sqrt{\frac{19}{24}} L$

7th Jan Morning**Sol****(1)**

$$\frac{ML^2}{12} + M\left(\frac{L}{4}\right)^2 = MK^2$$

$$\frac{L^2}{12} + \frac{L^2}{16} = K^2$$

$$K = \sqrt{\frac{7}{48}} L$$

mathongo

Q3

The surface mass density of a disc of radius a varies with radial distance as $\sigma = A + Br$ where A & B are positive constants then moment of inertia of the disc about an axis passing through its centre and perpendicular to the plane

एक चकती का सतही द्रव्यमान घनत्व त्रिज्जिय दूरी के अनुसार $\sigma = A + Br$ से परिवर्तित होता है, तथा चकती की त्रिज्या a है, तथा A और B धनात्मक नियतांक हैं, तो चकती के केन्द्र से तल के लम्बवत् गुज़रने वाली अक्ष के सापेक्ष जड़त्व आघूर्ण होगा

(1) $2\pi a^4 \left(\frac{A}{4} + \frac{Ba}{5} \right)$

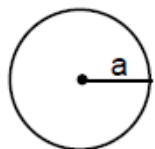
(2) $2\pi a^4 \left(\frac{Aa}{4} + \frac{B}{5} \right)$

(3) $\pi a^4 \left(\frac{A}{4} + \frac{Ba}{5} \right)$

(4) $2\pi a^4 \left(\frac{A}{5} + \frac{Ba}{4} \right)$

7th Jan Evening

(1)



$$\sigma = A + Br$$

$$\int dm = \int (A + Br) 2\pi r dr$$

$$I = \int dm r^2$$

$$= \int_0^a (A + Br) 2\pi r^3 dr$$

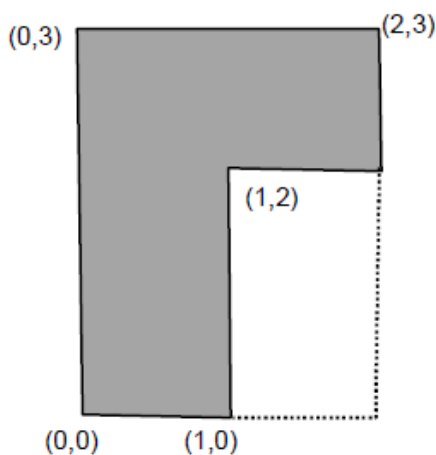
$$= 2\pi \left(A \frac{a^4}{4} + B \frac{a^5}{5} \right)$$

$$= 2\pi a^4 \left(\frac{A}{4} + \frac{Ba}{5} \right)$$

Q4

Find the co-ordinates of centre of mass of the lamina, shown in figure.

चित्र में दिखाई गई के प्लेट के द्रव्यमान केन्द्र के निर्देशांक होंगे :



(1) 0.75, 1.75

(2) 0.75, 1.5

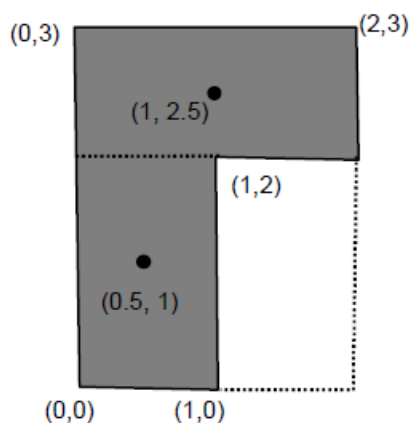
(3) 0.5, 1.75

(4) 0.5, 1.5

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Sol

(1)



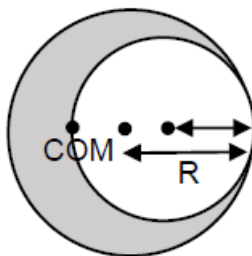
$$\vec{r}_{cm} = \frac{1 \times \left(\frac{\hat{i}}{2} + \hat{j} \right) + 1 \times \left(\hat{i} + \frac{5\hat{j}}{2} \right)}{2}$$

$$\vec{r}_{cm} = \frac{3}{4}\hat{i} + \frac{7}{4}\hat{j}$$

Q5

A uniform solid sphere of radius R has a cavity of radius 1m cut from it if centre of mass of the system lies at the periphery of the cavity then

एक समरूप R त्रिज्या के ठोस गोले में चित्रानुसार 1m त्रिज्या की गुहिका उपस्थित है इस निकाय का द्रव्यमान केन्द्र गुहिका की परिधि पर उपस्थित है तो :



(1) $(R^2 + R + 1)(2 - R) = 1$

(3) $(R^2 - R + 1)(2 - R) = 1$

(2) $(R^2 - R - 1)(2 - R) = 1$

(4) $(R^2 + R - 1)(2 - R) = 1$

 8th Jan Evening

Sol

(1)

$$M_1 = \frac{4}{3} \pi R^3 \rho$$

$$M_2 = \frac{4}{3} \pi (1)^3 (-\rho)$$

$$X_{\text{com}} = \frac{M_1 X_1 + M_2 X_2}{M_1 + M_2}$$

$$\Rightarrow \frac{\left[\frac{4}{3} \pi R^3 \rho \right] 0 + \left[\frac{4}{3} \pi (1)^3 (-\rho) \right] [R-1]}{\frac{4}{3} \pi R^3 \rho + \frac{4}{3} \pi (1)^3 (-\rho)} = -(2-R)$$

$$\Rightarrow \frac{(R-1)}{(R^3-1)} = (2-R) \quad (R \neq 1)$$

$$\frac{(R-1)}{(R-1)(R^2+R+1)} = 2-R$$

$$(R^2+R+1)(2-R) = 1$$

Alternative:

$$M_{\text{remaining}} (2-R) = M_{\text{cavity}} (1-R)$$

$$\Rightarrow (R^3-1^3)(2-R) = 1^3 [R-1]$$

$$\Rightarrow (R^2+R+1)(2-R) = 1$$

Q6

A solid sphere of mass $m = 500\text{gm}$ is rolling without slipping on a horizontal surface. Find kinetic energy of a sphere if velocity of centre of mass is 5 cm/sec .

$m = 500\text{gm}$ का एक ठोस गोला क्षेतिज सतह पर बिना घर्षण के लोटनी गति कर रहा है। यदि द्रव्यमान केन्द्र का वेग 5 cm/sec है, तो गोले की गतिज ऊर्जा ज्ञात किजिये।

$$(1) \frac{35}{2} \times 10^{-4} \text{ J}$$

$$(2) \frac{35}{4} \times 10^{-4} \text{ J}$$

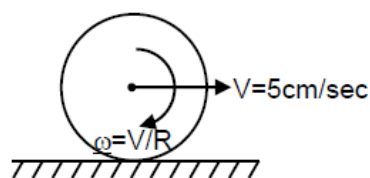
$$(3) 35 \times 10^{-4} \text{ J}$$

$$(4) 35 \times 10^{-3} \text{ J}$$

8th Jan Evening

Sol

(2)



K.E. of the sphere = Translational K.E + Rotational K.E.

गोले की गतिज ऊर्जा = स्थानांतरणिय K.E + घूर्णन K.E.

$$= \frac{1}{2}mv^2 \left(1 + \frac{K^2}{R^2} \right) \quad K = \text{Radius of gyration घूर्णन त्रिज्या}$$

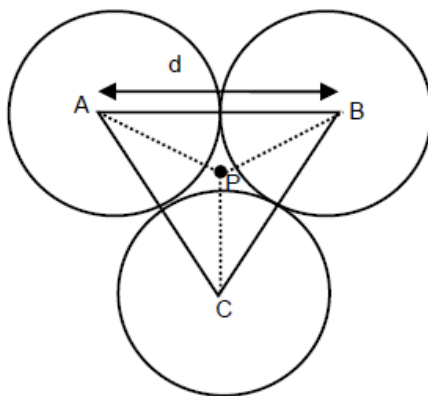
$$\frac{1}{2} \times \frac{1}{2} \times \left(\frac{5}{100} \right)^2 \left(1 + \frac{2}{5} \right)$$

$$\frac{35}{4} \times 10^{-4} \text{ J}$$

Q7

Three identical solid spheres each having mass 'm' and diameter 'd' are touching each other as shown in figure. Calculate ratio of moment of inertia about an axis (perpendicular to plane of paper) passing through point P and B as shown in figure. Given P is centroid of triangle ABC.

दिये गये चित्र अनुसार तीन एक समान ठोस गोले जिनका द्रव्यमान 'm' तथा व्यास 'd' है, को एक दूसरे के सम्पर्क में रखा जाता है। तो प्रदर्शित बिन्दु P तथा B से गुजरने वाली अक्ष (तल के लम्बवत्) के सापेक्ष जड़त्व आघूर्ण का अनुपात क्या होगा (दिया है : बिन्दु P, ΔABC का केन्द्रक है।)



(1) 13 / 23

(2) 11/19

(3) 7/9

(4) 13/11

9th Jan Morning

Sol

(1)

$$\text{M.I about P के परितः जड़त्व आघूर्ण} = 3 \left[\frac{2}{5} M \left(\frac{d}{2} \right)^2 + M \left(\frac{d}{\sqrt{3}} \right)^2 \right] = \frac{13}{10} M d^2$$

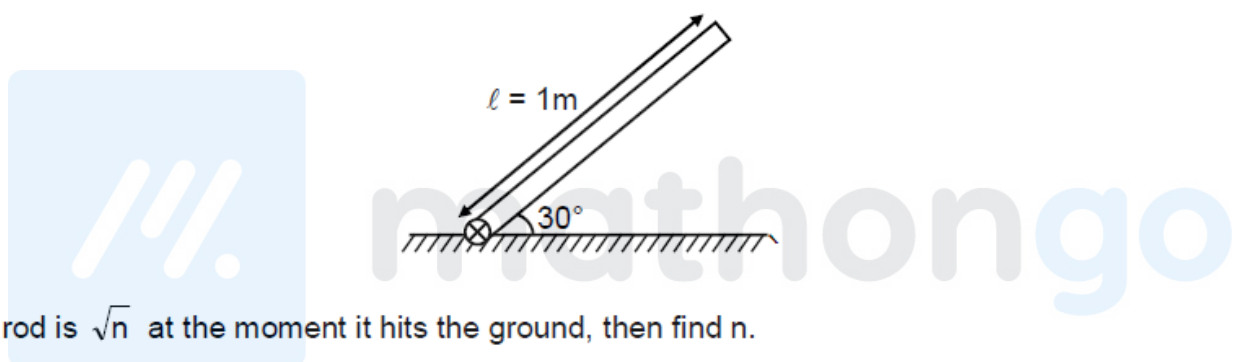
$$\text{M.I about B के परितः जड़त्व आघूर्ण} = 2 \left[\frac{2}{5} M \left(\frac{d}{2} \right)^2 + M (d)^2 \right] + \frac{2}{5} M \left(\frac{d}{2} \right)^2 = \frac{23}{10} M d^2$$

$$\text{Now ratio (अब अनुपात)} = \frac{13}{23}$$

Q8

A rod of length 1 m is released from rest as shown in the figure below.

दिये गये चित्रानुसार 1 m लम्बी छड़ को विराम से छोड़ा जाता है



If ω of rod is $\sqrt{n}$ at the moment it hits the ground, then find n.

9th Jan Morning

Sol

15

$$mg \frac{\ell}{2} \sin 30^\circ = \frac{1}{2} \frac{m \ell^2}{3} \omega^2$$

Solving

$$\omega^2 = 15$$

$$\omega = \sqrt{15}$$

Q9

Find centre of mass of given rod of linear mass density $\lambda = \left(a + b \left(\frac{x}{\ell} \right)^2 \right)$, x is distance from one of its end. Length of the rod is ℓ .

रेखीय द्रव्यमान घनत्व $\lambda = \left(a + b \left(\frac{x}{\ell} \right)^2 \right)$, जहाँ x एक सिरे से दूरी है, वाली छड़ का द्रव्यमान केन्द्र ज्ञात करो। ℓ छड़ की लम्बाई है।

- (1) $\frac{3\ell}{4} \left(\frac{3a+b}{2a+b} \right)$ (2) $\frac{3\ell}{4} \left(\frac{2a+b}{3a+b} \right)$ (3) $\frac{\ell}{4} \left(\frac{2a+b}{3a+b} \right)$ (4) $\ell \left(\frac{2a+b}{3a+b} \right)$

9th Jan Evening

Sol

(2)



$$x_{cm} = \frac{1}{M} \int_0^{\ell} x \cdot dM$$

$$dM = \lambda \cdot dx = \left(a + b \left(\frac{x}{\ell} \right)^2 \right) \cdot dx$$

$$x_{cm} = \frac{\int x dM}{\int dM} = \frac{\int x \lambda dx}{\int \lambda dx} = \frac{\int_0^{\ell} x \left(a + \frac{bx^2}{\ell^2} \right) dx}{\int_0^{\ell} \left(a + \frac{bx^2}{\ell^2} \right) dx}$$

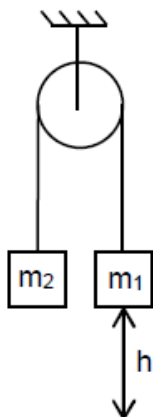
$$= \frac{a \left(\frac{x^2}{2} \right)_0^{\ell} + \frac{b}{\ell^2} \left(\frac{x^4}{4} \right)_0^{\ell}}{a(x)_0^{\ell} + \frac{b}{\ell^2} \left(\frac{x^3}{3} \right)_0^{\ell}}$$

$$= \frac{a \frac{\ell^2}{2} + b \frac{\ell^2}{4}}{a\ell + \frac{b\ell}{3}} = \frac{(2a+b) \ell}{(3a+b) 4} \times 3$$

$$= \frac{3\ell}{4} \left(\frac{2a+b}{3a+b} \right)$$

System is released from rest. Moment of inertia of pulley 'I'. Find angular speed of pulley when m_1 block falls by 'h'. (Given $m_1 > m_2$ and assume no slipping between string and pulley).

'I' जड़त्व आघूर्ण वाली धिरनी का कोणीय वेग ज्ञात करो, जब एक ब्लॉक 'h' दूरी से नीचे चला जाता है। ($m_1 > m_2$) तथा धिरनी व रस्सी के मध्य कोई फिसलन नहीं है।



(1) $\frac{1}{R} \sqrt{\frac{2(m_2 - m_1)gh}{\left(m_1 + m_2 + \frac{I}{R^2}\right)}}$

(2) $\frac{1}{R} \sqrt{\frac{2(m_2 + m_1)gh}{\left(m_1 + m_2 + \frac{I}{R^2}\right)}}$

(3) $\frac{1}{R} \sqrt{\frac{(m_2 - m_1)gh}{\left(m_1 + m_2 + \frac{I}{R^2}\right)}}$

(4) $\frac{1}{R} \sqrt{\frac{(m_2 + m_1)gh}{\left(m_1 + m_2 + \frac{I}{R^2}\right)}}$

9th Jan Evening

Sol

(1)

$$K_i + U_i = K_f + U_f$$

$$0 + 0 = \frac{1}{2}m_2v^2 + \frac{1}{2}m_1v^2 + \frac{1}{2}I\omega^2 - m_2gh + m_1gh$$

$$(m_2 - m_1)gh = \frac{1}{2}m_2(\omega R)^2 + \frac{1}{2}m_1(\omega R)^2 + \frac{1}{2}I\omega^2$$

$$\sqrt{\frac{2(m_2 - m_1)gh}{\left(m_1 + m_2 + \frac{I}{R^2}\right)R^2}} = \omega$$

$$\frac{1}{R} \sqrt{\frac{2(m_2 - m_1)gh}{\left(m_1 + m_2 + \frac{I}{R^2}\right)}}$$

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